# LaTeX Installation Test

Neovim User

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## 1 Introduction

If you are reading this PDF, your $\LaTeX$ installation works! This document tests basic compilation and math rendering.

## 2 Math Test

Here is an example of a mathematical equation to verify that the `amsmath` package is installed correctly:

$$E = mc^2 \tag{1}$$

And an integral:

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$$

Ready for Neovim editing!

# 3 Abstract

This project focuses on the further development of a method for offline tuning of rate controllers for FPV racing drones based on the open-source firmware Betaflight. The quality of rate control is crucial for flight stability, suppression of disturbances such as prop wash, and the agility of the drone. Until now, tuning has mostly been done through time-consuming trial and error.

The aim of the project is to develop a modular, open-source library for offline tuning based on an existing proof of concept. The library will be implemented in MATLAB and a light version in Python and supplemented by detailed Markdown documentation with examples on GitHub. The solution should offer the community a simple way to efficiently optimise drone parameters and thereby improve flight performance.

The results will be published in a public GitHub repository and documented in a report. In addition to software development, the project also includes practical work with FPV drones to validate the methods developed.

# 4 Foreword

This document shows the results of our project thesis, which we worked on during the Fall Semester 2025 at the ZHAW School of Engineering. For us as Systems Engineering students, the Betaflight project was the perfect opportunity to apply complex control theory and signal processing concepts to a real, highly dynamic system. While FPV drones originally started as a hobby for one of our team members, we all became fascinated by the engineering challenge of mathematically modelling and optimising their flight behaviour.

Our motivation was to bridge the gap between academic theory and practical application. We aimed to develop a tool that not only meets engineering standards but also provides real value to the FPV community by replacing subjective tuning with objective data analysis.

This work was carried out at the Institute of Mechatronic Systems (IMS). We would also like to thank our supervisors, Michael Peter and also Prof. Dr. Ruprecht Altenburger, who was our assistant supervisor. They helped us with the technical side of our project and gave us the equipment we needed.

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# 5 Introduction

First-person view (FPV) racing drones require precise and responsive control to achieve high flight stability, minimize disturbances such as prop wash, and maintain agility during rapid maneuvers. The tuning of rate controllers is key to this performance, as it's what determines how the drone responds to pilot inputs and external influences. Until now, tuning these controllers has relied on iterative trial-and-error methods, a process that is both time-consuming and really hard to do for less experienced pilots.

Betaflight, an open-source firmware widely used in the FPV community, provides a flexible platform for configuring flight controllers. However, despite its popularity, efficient tuning remains a challenge. To address this, our project builds upon an existing proof of concept for offline tuning of rate controllers using flight data and system identification techniques. The goal is to transform this concept into a robust, modular, and open-source library that simplifies the tuning process.

The library will be implemented in 'MATLAB and Python', offering tools for analyzing flight logs, estimating system dynamics, and optimizing controller parameters without requiring repeated flight tests. A documentation is created, which shows how to use the tool and will be published on GitHub to ensure accessibility for the FPV community.

Besides software development, the project involves also building an FPV drone and testing and confirm the effectiveness of the proposed methods. By providing a structured, data-driven approach to tuning, this work aims to improve flight performance and reduce the barriers to achieving optimal control settings for both hobbyists and professionals.